

Certified Penetration Testing Professional

Methodology: OT and SCADA Penetration Testing

Penetration Tester:			
Organization:			
Date:		**Location:**	

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Test 1: Modbus

Test 1.1: Modbus Recon

Target Organization		
URL		
Target OT System		
Commands Used		
Modus Recon Successful?	☐ Yes	☐ No
Modbus Protocol Running on Port		
Identified Station IDs		
Other Information Gathered		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 2: ICS and SCADA Pen Testing

Test 2.1: ICS and SCADA Pen Testing

Target Organization	
URL	
Target OT System	
Step-1: Asset and System Identification	
List of Components Allowed for Testing	1. 2. 3. 4. 5. 6. 7.
List of Components Prohibited from Testing	1. 2. 3. 4. 5. 6. 7.
Method Used for Testing	1. 2. 3. 4. 5.

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Identified Threat Vectors			
Threat Vector		**Priority**	**Description**
1.			
2.			
3.			
4.			
5.			

Step-2: Vulnerability Identification	
Techniques Used for Vulnerability Identification	1. 2. 3. 4. 5. 6. 7.

Identified Vulnerabilities				
Vulnerability		**Attack Surfaces**	**Threat Vector**	**Risk Level**
1.				
2.				
3.				
4.				
5.				

Step-3: Risk Determination			
Vulnerability		**Risk Level**	**Impact Caused**
1.			
2.			
3.			
4.			
5.			

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Other Information Obtained through Pen Testing	1. 2. 3. 4. 5.
Tools/Services Used	1. 2. 3. 4. 5.

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Test 2.2: Attack Modifications

Target Organization	
URL	
Target OT System	
List of Components Allowed for Testing	1. 2. 3. 4. 5.
List of Components Prohibited from Testing	1. 2. 3. 4. 5. 6.
Method of Attacking Site	
List of Data Required to Conduct the Test	1. 2. 3. 4. 5. 6.

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Testing Plan	
Recovery Plan	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 2.3: Test OT using Different Tools

Target Organization		
URL		
Target OT System		
Commands Used		
Testing of OT Successful?	☐ Yes	☐ No

Information Obtained				
	Target IP Address	**Live Hosts/Devices**	**Configuration of Devices**	**Vulnerabilities Identified**
1.				
2.				
3.				
4.				
5.				
6.				
7.				
8.				
9.				
10.				
11.				
12.				
13.				
14.				
15.				

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Tools/Services Used	1. 2. 3. 4. 5.

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Test 2.4: Port Scanning

Target Organization					
URL					
Target OT System					
Low-risk Scan					
Commands Used					
	Device Scanned	**IP Address**	**Running Services**	**Identified Ports**	**Port State (Open/ Closed/Filtered)**
1.					
2.					
3.					
4.					
5.					
Other Information Obtained through Scanning	1.				
	2.				
	3.				
	4.				
	5.				
Medium-risk Scan					
Commands Used					

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	Device Scanned	IP Address	Running Services	Identified Ports	Port State (Open/ Closed/Filtered)
1.					
2.					
3.					
4.					
5.					
Other Information Obtained through Scanning	1. 2. 3. 4. 5.				
Tools/Services Used	1. 2. 3. 4. 5.				

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Test 2.5: Vulnerability Scanning using Nessus

Target Organization			
URL			
Target OT System			
Commands Used			
Vulnerability Scanning using Nessus Successful?	☐ Yes	☐ No	
Created New Scan Profile?	☐ Yes	☐ No	
Disabled Nessus TCP, SYN, UDP, and SNMP Port Scans?	☐ Yes	☐ No	
All Plugins are disabled Expect for Windows/Linux compliance checks?	☐ Yes	☐ No	
Nessus Profile is Updated?	☐ Yes	☐ NO	
Vulnerabilities Identified	1. _____ 2. _____ 3. _____ 4. _____ 5. _____		
Tools/Services Used	1. _____ 2. _____ 3. _____ 4. _____ 5. _____		

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